

Temporal Predictions with NN RNN and LSTM

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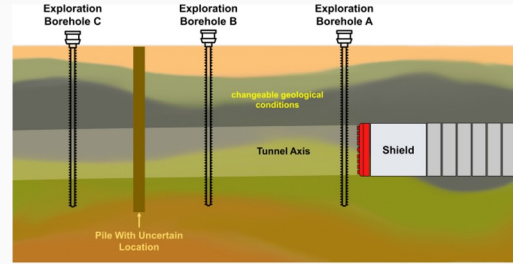
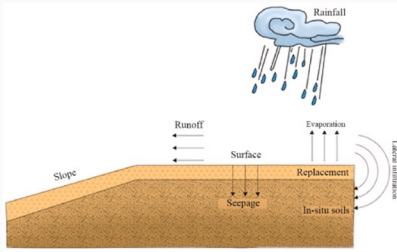
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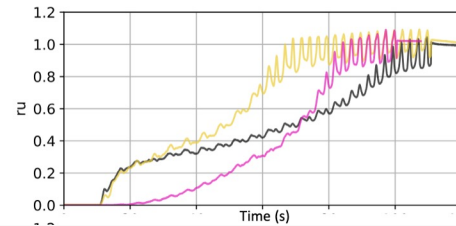
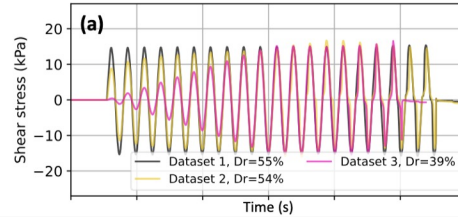
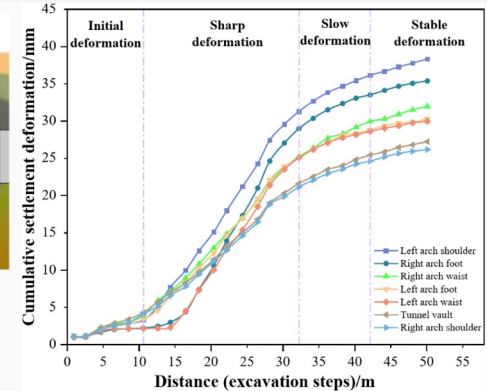
Learning Objectives

- Understand the need for temporal predictions
- Why MLPs can't work for temporal predictions (need for history)
- Implement RNN/LSTM models for geotechnical time-series prediction (settlement, pore pressure, displacement)

Temporal Predictions

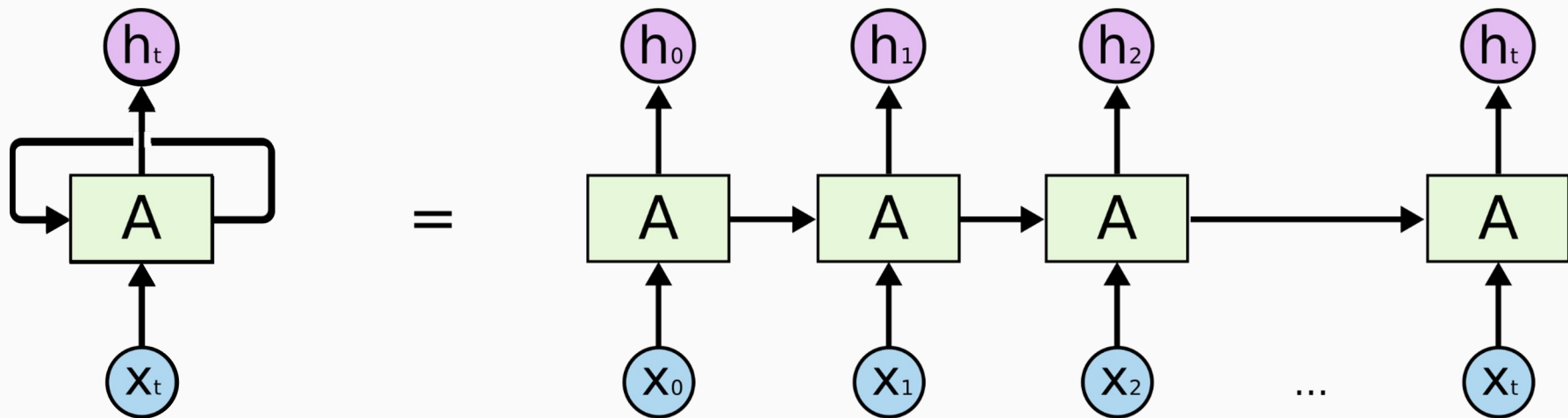


<https://www.nature.com/articles/s41598-024-79059-x>



<https://doi.org/10.1061/9780784484692.021>

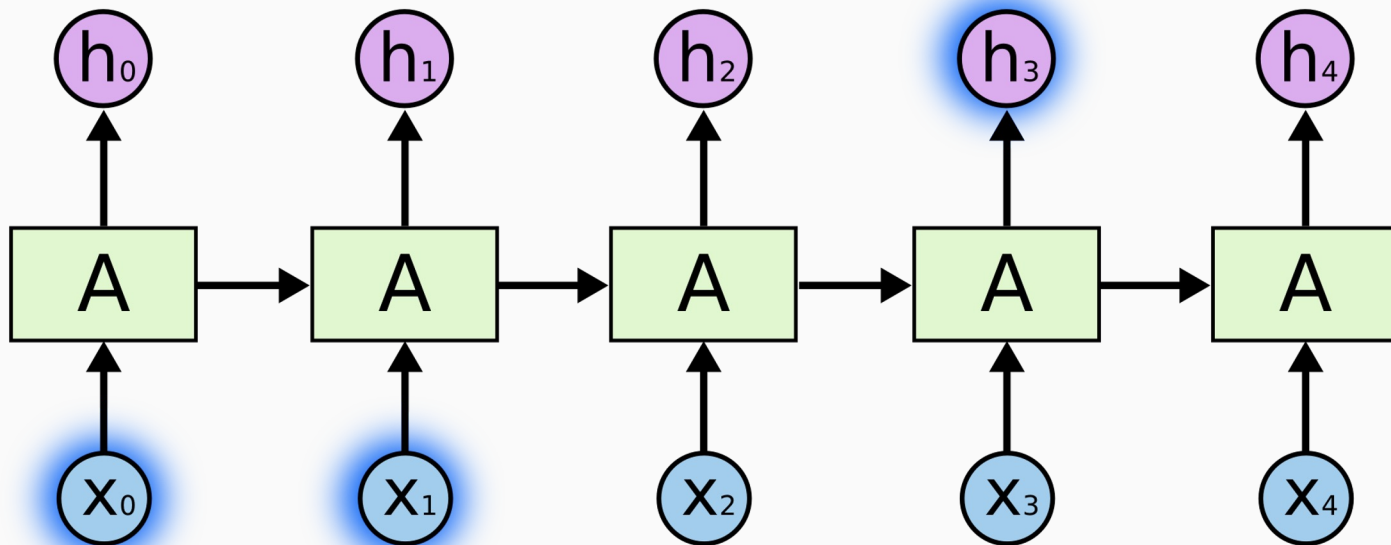
Remembering the past...Recurrent Networks



Recurrent Neural Networks

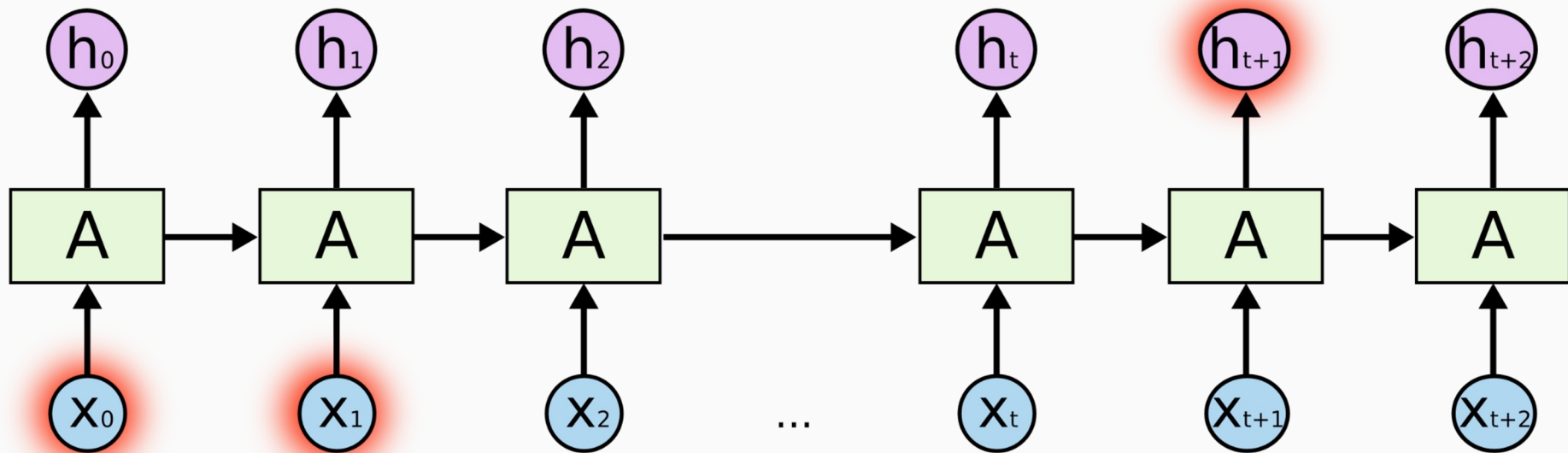
Predicting based on the past...

the clouds are in the *sky*

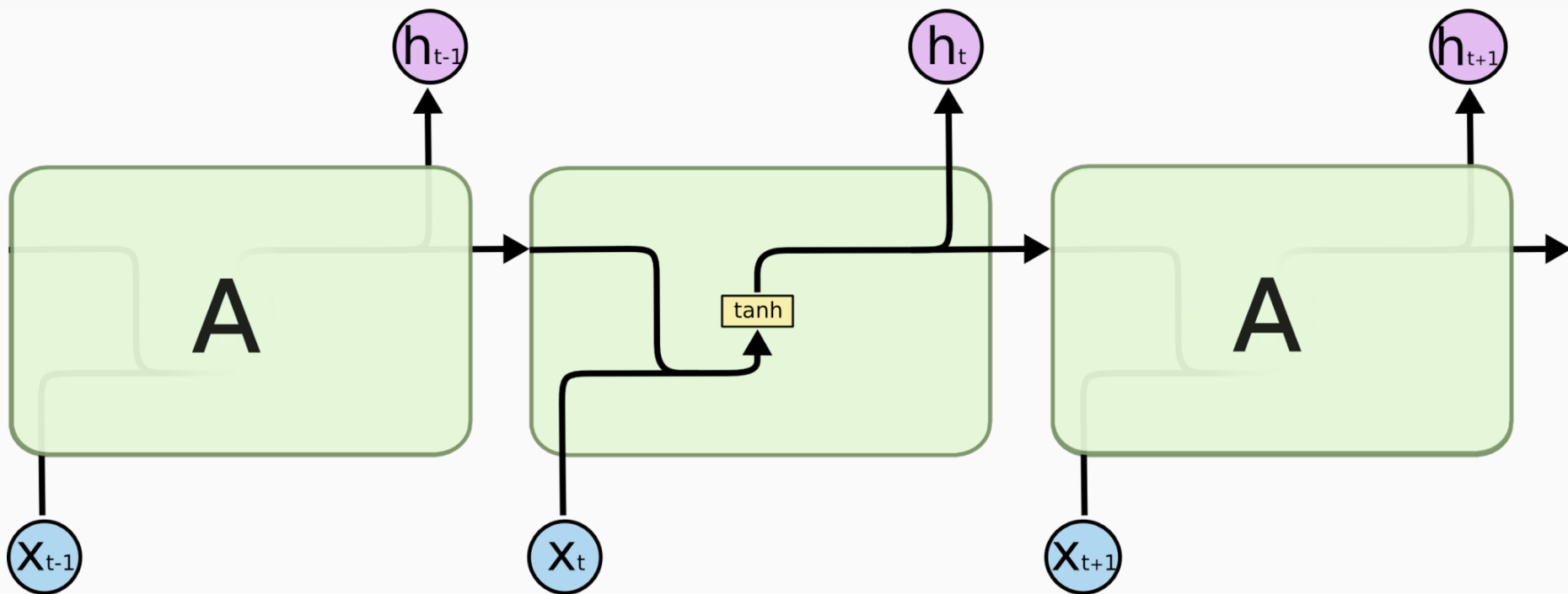


The problem of long dependencies

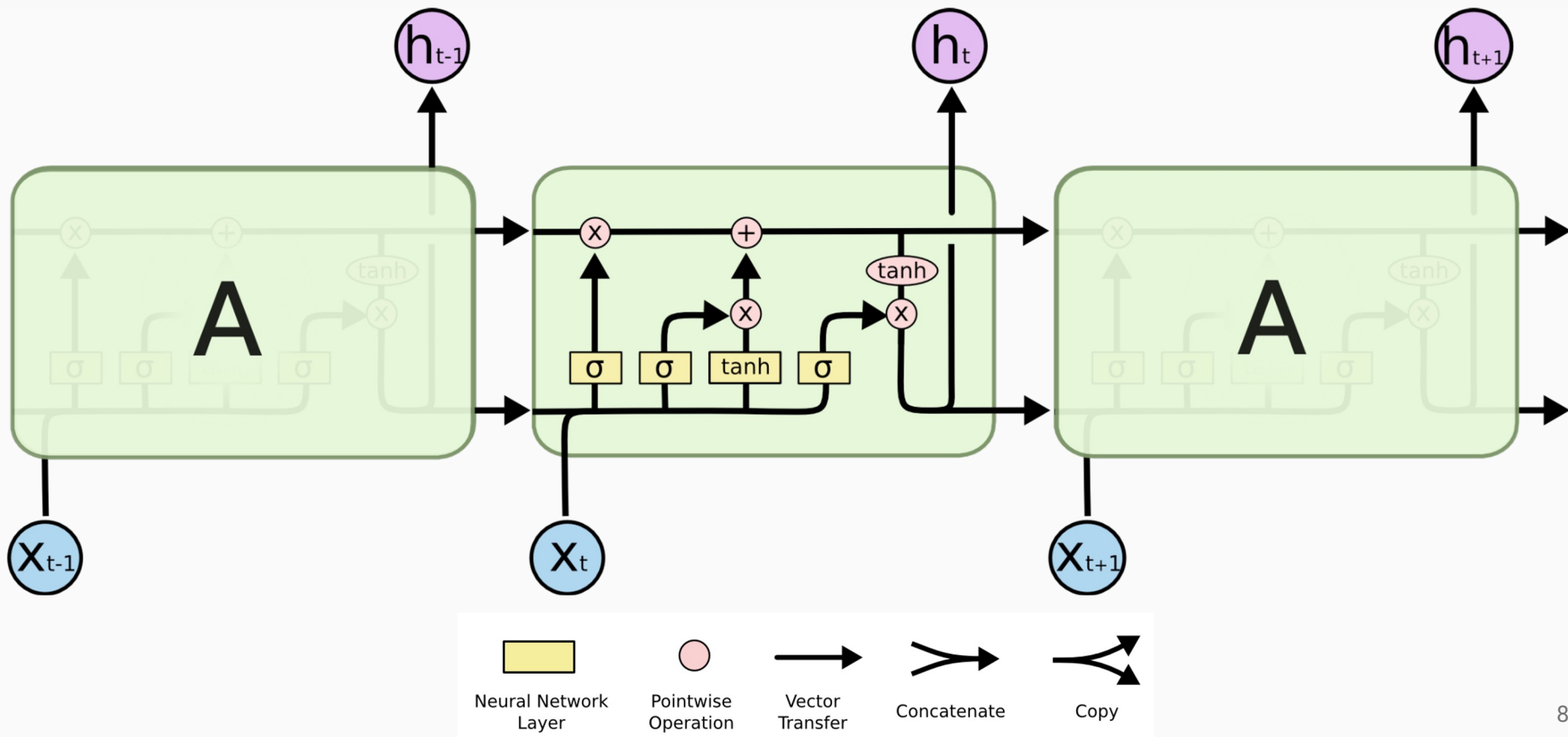
I grew up in France... Although I live in the US, now, I speak fluent ***French***



Long Short Term Memory (LSTM)



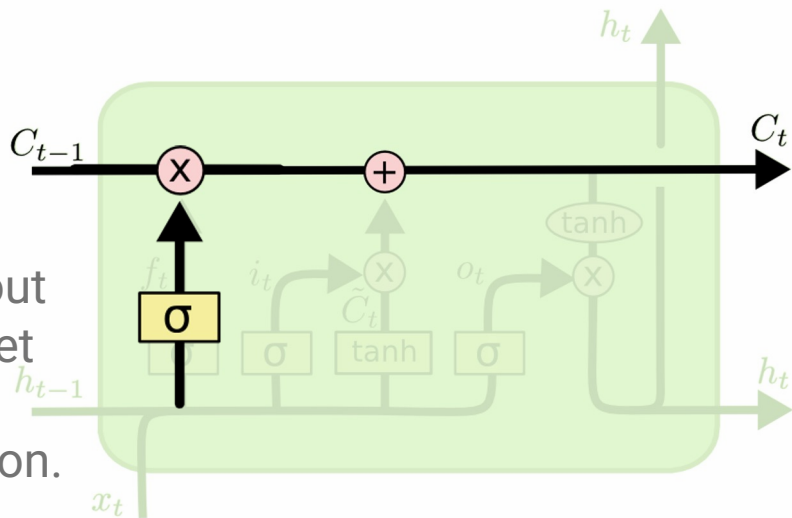
LSTM components



Core Idea: LSTM – Cell State

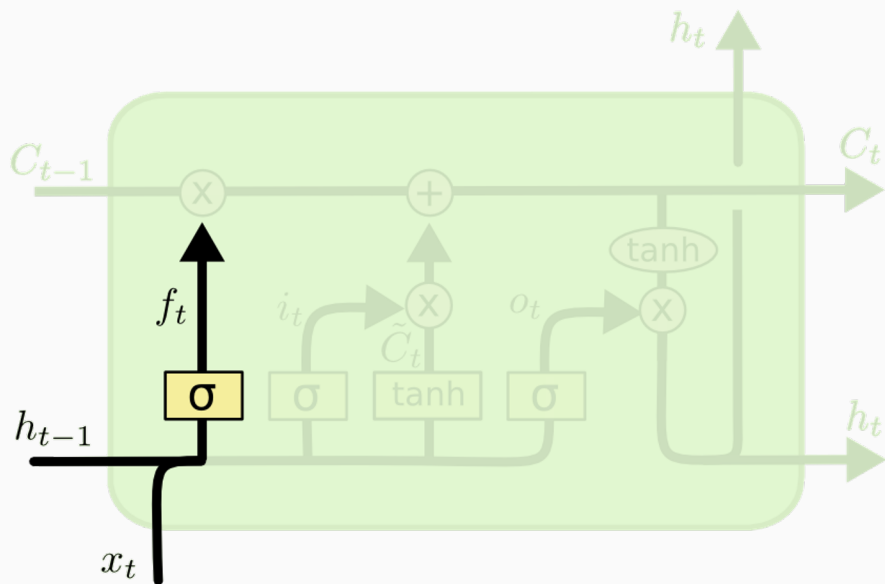
The cell state is kind of like a conveyor belt. Runs straight down the entire chain, with only some minor linear interactions. LSTM does have the ability to remove or add information to the cell state, carefully regulated by structures called gates

Gates are a way to optionally let information through. They are composed out of a sigmoid neural net layer and a pointwise multiplication operation.



LSTM – Step-by-Step (Forget Gate)

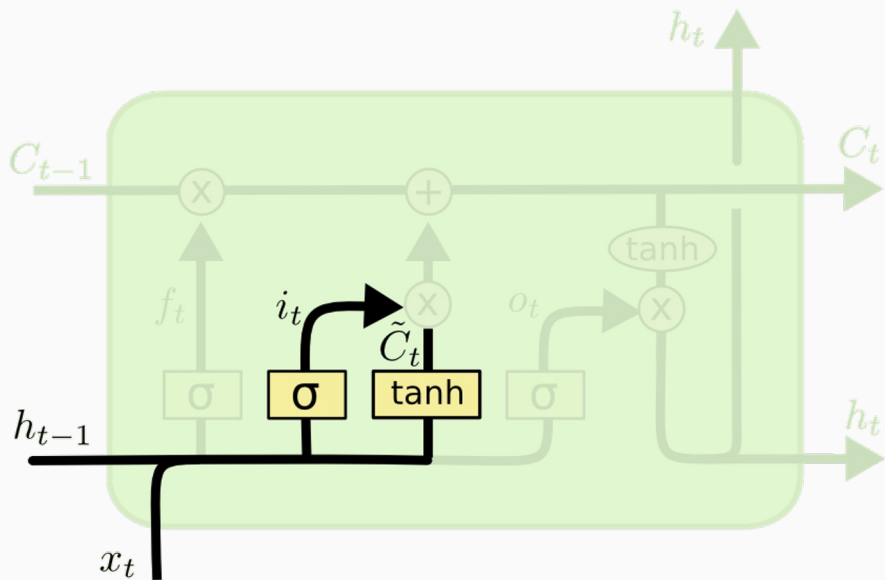
what information we're going to throw away from the cell state



$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$

LSTM – Step-by-Step (Input Gate)

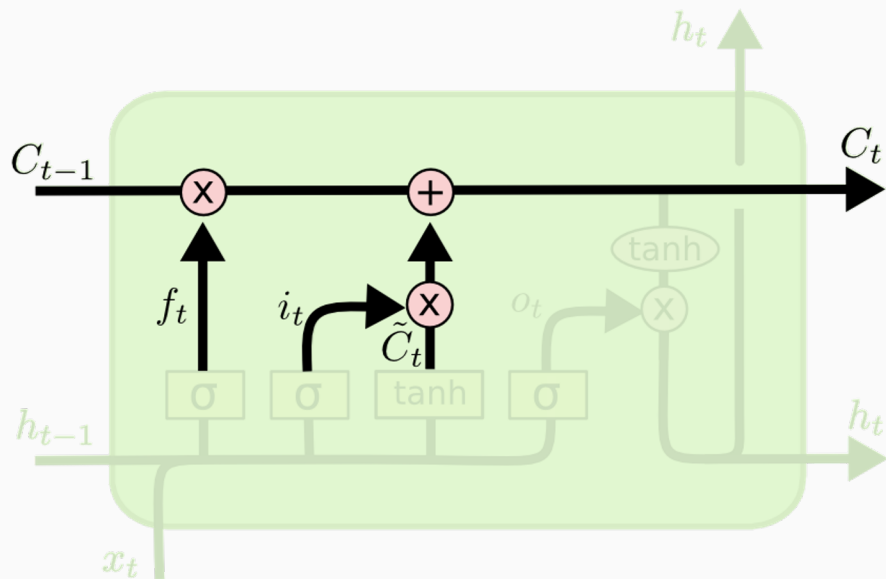
what new information we're going to store in the cell state... a sigmoid layer called the "input gate layer" decides which values we'll update. Next, a tanh layer creates a vector of new candidate values, that could be added to the state



$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

LSTM – Step-by-Step (Update Old Cell State)

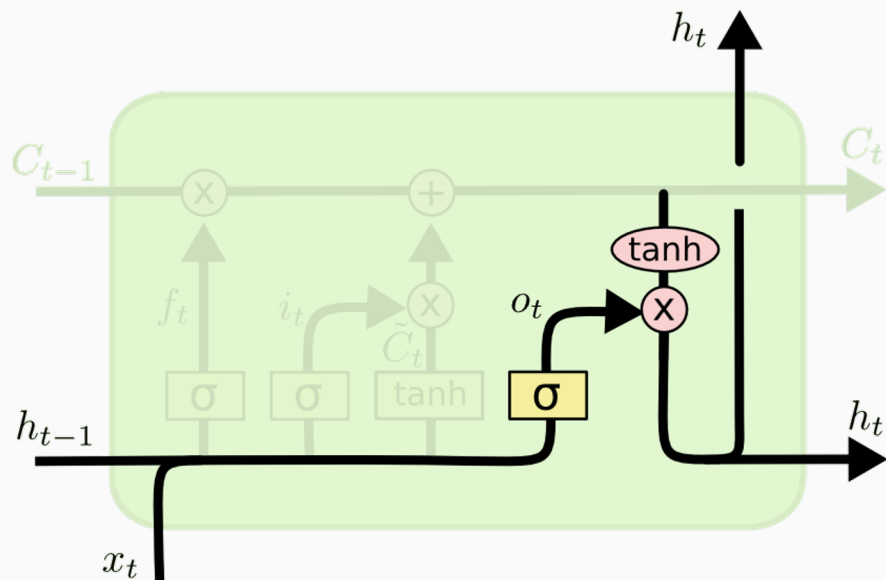
Update old cell state C_{t-1} to C_t



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

LSTM – Step-by-Step (Output)

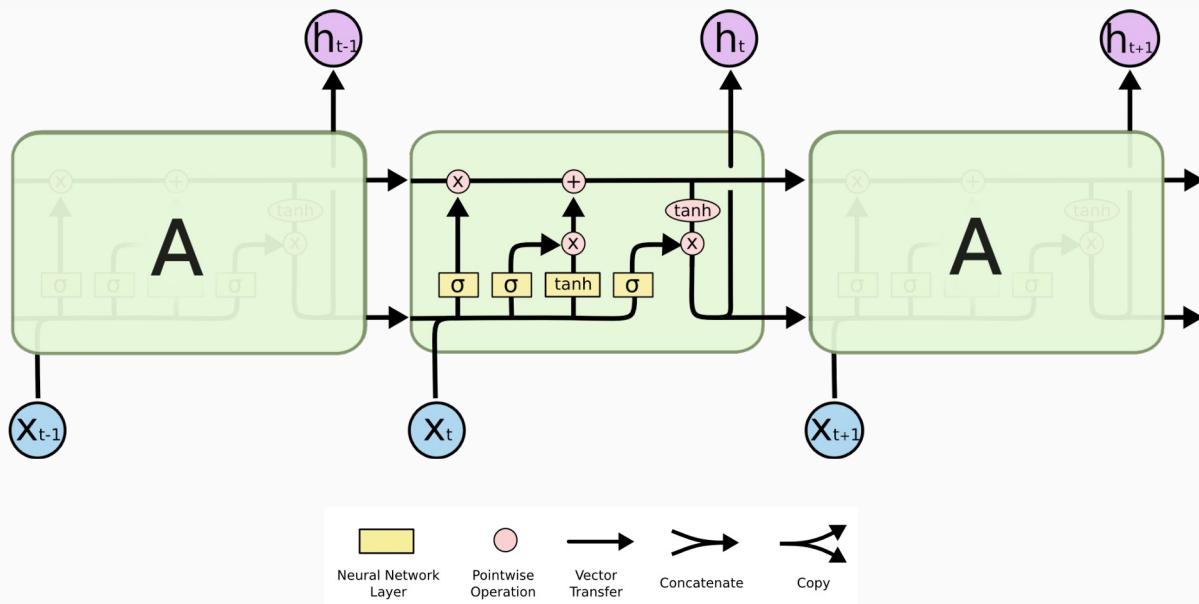
what we're going to output. This output will be based on our cell state, but will be a filtered version



$$o_t = \sigma (W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh (C_t)$$

LSTM Network



$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$$

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$$

$$\tilde{C}_t = \tanh(W_{xc} \times x_t + W_{hc} \times h_{t-1} + b_c)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

$$h_t = o_t \odot \tanh(C_t)$$

LSTM Prediction of Liquefaction

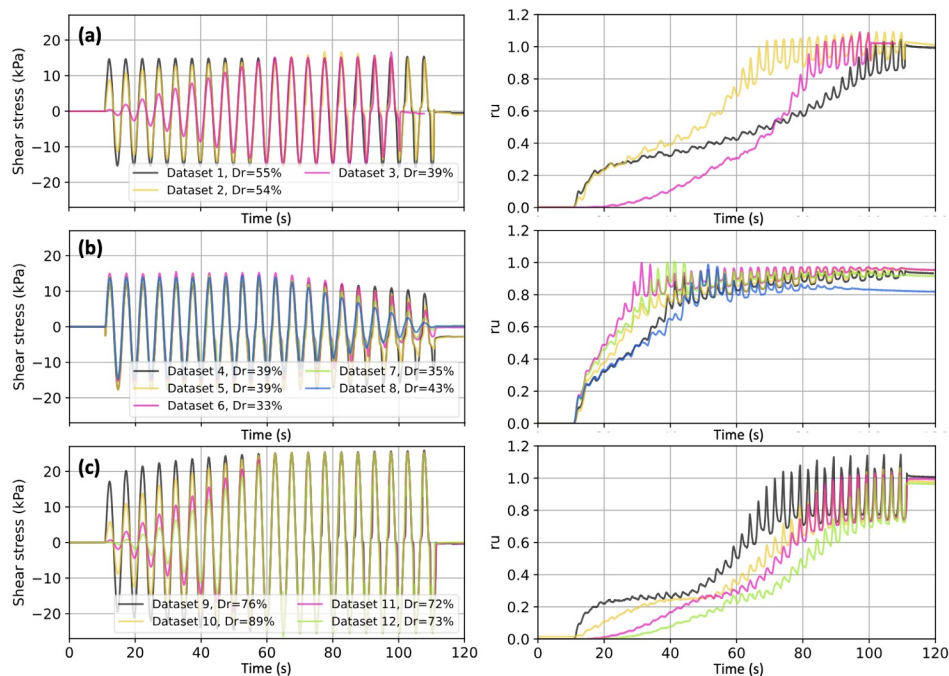


Figure 3. The responses of pore water pressure (r_u) corresponding to different input shear stresses and sample relative densities (D_r) for 12 laboratory cyclic simple shear tests (Kwan et al., 2017): (a) one harmonic and two modulated-up stresses on loose samples; (b) five modulated-down stresses on loose samples; (c) four modulated-up stresses on dense samples.

LSTM Cell

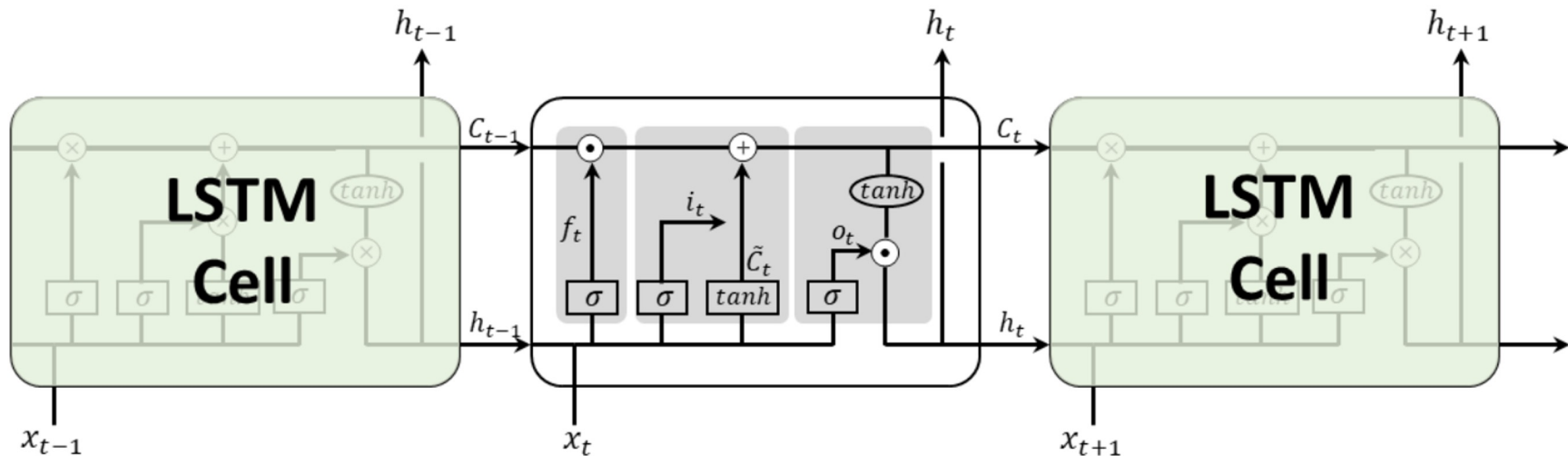
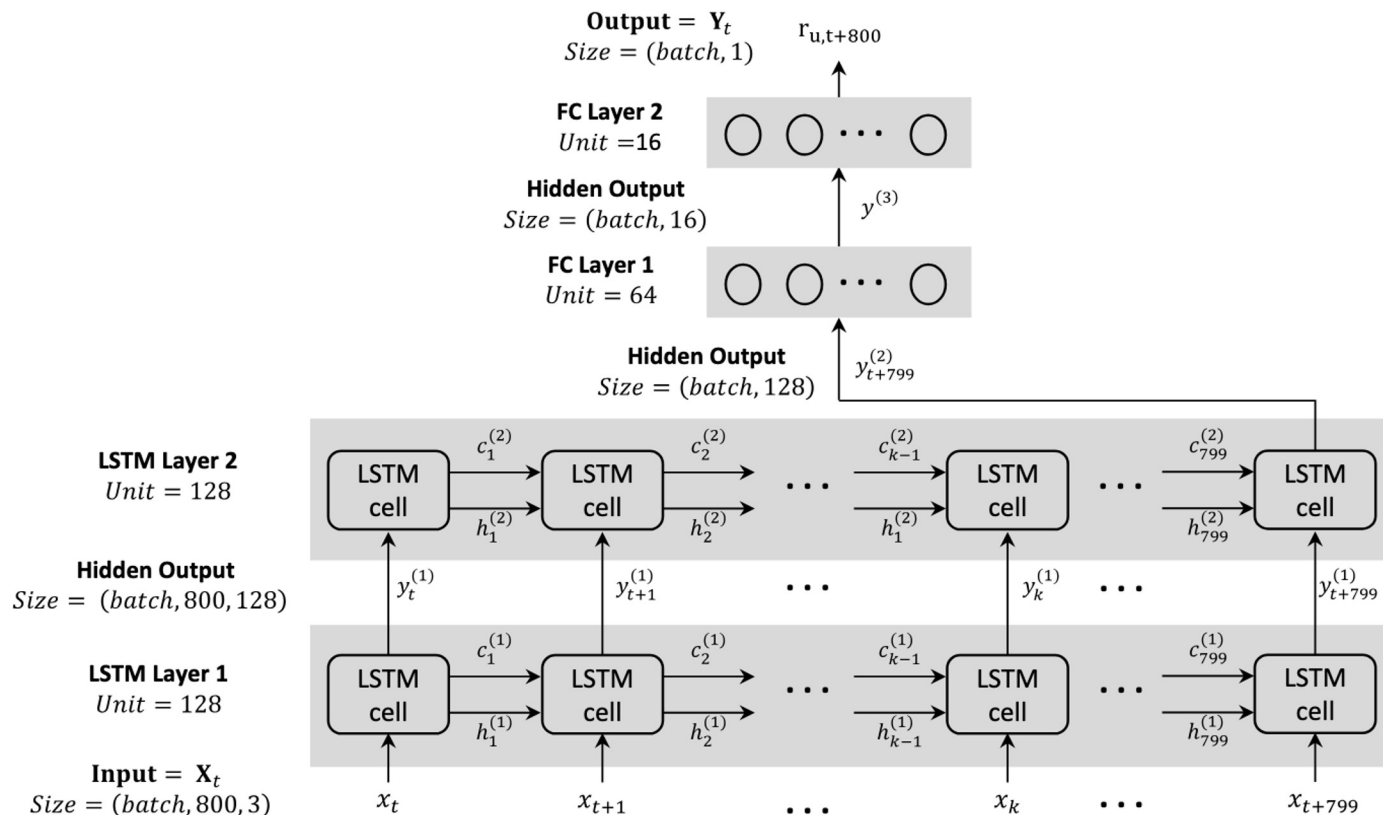


Figure 2. LSTM cell architecture. The subscript t stands for the time step. x , h , and c are the input, hidden output, and cell state. f_t , i_t , and o_t are the forget, input, and output gate at t . σ and $\tanh$ are the sigmoid and hyperbolic tangent function. $\odot$ and $\oplus$ are the element-wise product and summation.

LSTM Network



LSTM Network predictions

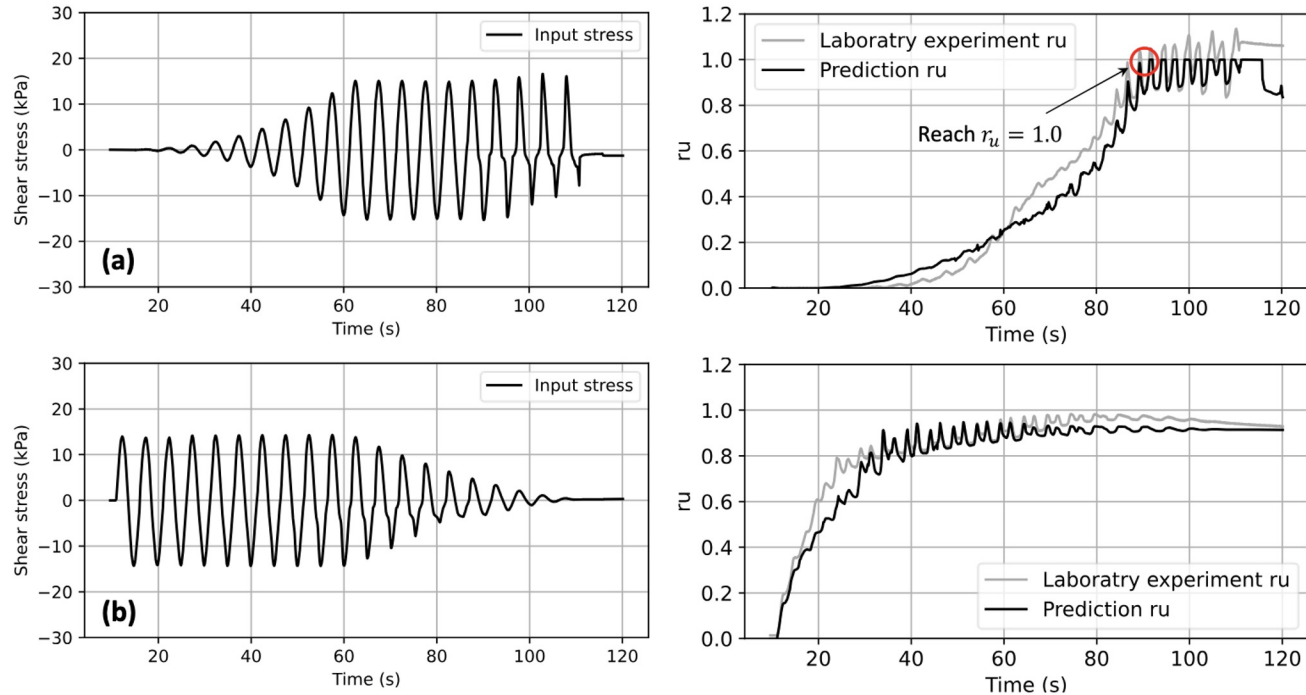


Figure 6. LSTM prediction of pore pressure response compared with laboratory experiment: (a) Loose sample with $D_r=41\%$ and modulated-up loading and (b) Loose sample with $D_r=35\%$ and modulated-down loading.